HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and refined **Chain-of-Thought (CoT) reasoning** for complex coding tasks across **Python** and **Java**, improving AI's logical accuracy in algorithmic problem-solving.
- Audited AI-generated **Python** and **JavaScript** code for adherence to **clean-code principles** (SOLID, DRY) and identified potential **security vulnerabilities** such as SQL injection.
- Designed and executed multi-turn debugging scenarios to test AI's ability to identify, explain, and fix bugs in provided code blocks, enhancing its diagnostic capabilities.
- Optimized the translation of **natural language prompts** into functional **Java** and **Python** code outputs, ensuring precise capture of subtle logic requirements for AI-driven code generation.
- Executed and analyzed performance benchmarks for **coding models**, identifying and documenting "logic gaps" in handling specific **Python libraries** and frameworks.
- Containerized **Java Spring Boot** applications and deployed them to **AWS EKS**, managing basic resource limits and configurations for predictable rollouts.
- Mentored junior engineers on effective code review practices and adherence to **SOLID principles** within an Agile development team, improving overall code quality.

SKILLS

AI/ML – Large Language Models (LLMs), Chain-of-Thought (CoT), NLP, Programmatic Reasoning, Algorithmic Reasoning, Benchmarking, Logic Gaps, Fine-tuning

Languages – Python, Java, JavaScript, C++, SQL

Software Development – Clean Code, SOLID Principles, DRY Principles, Code Quality, Security Auditing, Debugging, Code Generation, REST APIs

Cloud & DevOps – AWS, EKS, Docker, CI/CD, Jenkins, Git

Frameworks/Libraries – Spring Boot, scikit-learn, pandas, NumPy, Java EE, JUnit

PROJECTS

AI Code Generation & Reasoning Engine (HCL Technologies)

- Developed a **Python**-based engine to translate complex natural language requirements into functional **Java** code, achieving a **78% success rate** on initial logic translation for internal tools.
- Implemented **Chain-of-Thought (CoT) prompting** strategies to guide an LLM in solving algorithmic challenges, improving its step-by-step reasoning accuracy by approximately **15%** on benchmark datasets.
- Integrated **Docker** containers for deploying and testing AI-generated code snippets in isolated environments, ensuring consistency across development and validation phases.

LLM Code Quality & Security Auditor (MSc Applied Project)

- Built a **Python** analysis tool that audits AI-generated code for adherence to **SOLID principles** and flags potential **security vulnerabilities** like buffer overflows and injection flaws.
- Evaluated the tool's effectiveness by processing over 500 code samples, identifying an average of **3 critical security flaws** per batch of 100 generated snippets.
- Designed a **multi-turn debugging strategy** simulation to assess LLM's ability to fix identified code quality and security issues interactively.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)